Minimum-Rank and Mortality Bounds for Finite Real Matrix Monoids

github.com/egilburg/aimath

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Abstract

For a family of real $n \times n$ matrices whose entire product monoid is finite, let s be its minimum rank. We prove that a word attaining rank s has length at most

$$B(n,s) = n2^{n-s} - \frac{n(n+1)}{2} + \frac{s(s-1)}{2}.$$

In the mortal case this gives $B(n,0) = \Theta(n2^n)$, improving the $\Theta(n^2 2^n)$ bound in revision r2 of this manuscript. Over the rationals, it improves Kiefer and Ryzhikov's (2026) bound 3^{n^2} for both mortality and minimum-rank diameter, and Almeida and Steinberg's (2009) universal mortality bound $(2n-1)^{n^2} - 1$ for $n > 1$. We also prove the sharp universal mortality threshold four in dimension two, additive bounds along invariant flags with sharp examples for factors of dimensions one and two, and existence of a minimum-rank witness with at most $n^3 + n^2 + 3n + 1$ straight-line-program gates. A bounded, individually periodic pair on the real plane shows why finiteness of the entire monoid matters. These statements have accompanying Lean proofs; the compressed-witness result asserts existence, without a polynomial-time synthesis claim. Discoveries presented here were made through the use of artificial intelligence.

1 Scope, bounds and prior work

For an arbitrary indexing set A, let $M_a \in M_n(\mathbb{R})$ and put

$$M_{a_1 \cdots a_m} = M_{a_1} \cdots M_{a_m}, \qquad M_\epsilon = I_n, \qquad S = \{M_w : w \in A^*\}.$$

Products are written in list order; on column vectors the rightmost matrix acts first. Throughout the finite-monoid results, *the entire set S is finite*. The indexing set need not be finite. Mortality means $0 \in S$. The minimum rank is $s = \min_{X \in S} \operatorname{rank} X$.

Theorem 1.1. *For $n \geq 0$, a finite monoid S as above has a word w with*

$$\operatorname{rank} M_w = s, \qquad |w| \leq B(n,s) = n2^{n-s} - \frac{n(n+1)}{2} + \frac{s(s-1)}{2}.$$

The same assertion holds over $\mathbb{Q}$. In particular, if S is mortal then a zero word has length at most $B(n,0)$. If $s = n$, the empty word suffices. For $n = 0$, the empty matrix is both identity and zero and the bound is zero.

The expression is nonnegative for $0 \leq s \leq n$. Its Lean definition adds the nonnegative terms before natural-number subtraction; Section 4 proves an additive identity justifying the displayed integer formula.

1.1 What changes from the previous edition

Revision r2 [6] proved only the mortality bound

$$b(n) = 2^{n-1} + (2^{n-1} - 1)\frac{n(n+1)}{2} \quad (n \geq 1).$$

The new bound is strictly smaller for $n \geq 3$, with the exact comparison

$$b(n) - B(n, 0) = 2^{n-2}(n-1)(n-2) \quad (n \geq 2). \tag{1}$$

Both bounds give one at $n = 1$ and five at $n = 2$; Section 5 improves the latter to the optimal four. For example, at dimensions three and four the old bounds $22, 78$ become $18, 54$. The positive minimum-rank case, sharp factor examples, boundary example and compressed witnesses are also additions. The present general upper bound is not claimed to be sharp.

1.2 Matched comparisons and attribution

Almeida–Steinberg [1, Definitions 2.1 and 5.1, Theorem 5.6] give a universal rational mortality function equal to one at dimension one and $(2n-1)^{n^2} - 1$ above it. Applying Theorem 1.1 to the image of any finite-monoid representation supplies another such function. Its exponential scale already occurred in r2; this edition improves its polynomial factor. This addresses the exponential alternative of their Question 5.7, without settling the polynomial alternative or claiming historical priority for its resolution.

Kiefer–Ryzhikov [4, Theorem 7.1 and Proposition 7.3] bound mortality and minimum-rank diameter by 3^{n^2} for finite rational semigroups, with no irreducibility hypothesis on these two results. Our $B(n, s)$ improves these particular length bounds and extends the field to $\mathbb{R}$. Their irreducible cardinality theorem is a different result. Adjoining identity aligns semigroup and monoid conventions; if the minimum rank is full and nonempty words are required, a generator itself has full rank and length one. Otherwise the word supplied here is automatically nonempty.

The ingredients have substantial antecedents. Finite-group averaging and symmetric-matrix lifting, including dimension $n(n+1)/2$, occur in Protasov–Voynov [7, Sections 7 and 9.1]. Almeida–Steinberg's Lemma 2.4 and Proposition 2.5 already combine annihilators along invariant subspaces; Section 6 uses that mechanism with the bounds proved here. The plane lower example comes from the classical Černý construction [2], not a new extremal construction. Our quantitative refinement uses a *matrix-valued* invariant-form defect of known codimension and a rank-sensitive sandwich recurrence.

On nonnegative matrices, $2^n - 1$ is a stronger mortality bound without the finite arithmetic-monoid promise [8]. Compressed minimum-rank witnesses and algorithms also exist for monoids consisting entirely of zero–one matrices [3]. Our signed real/rational statement does not replace those more specialized algorithmic results. We make no first-or-best assertion across all literature. Neither a polynomial universal mortality bound nor a polynomial-time algorithm for the general signed input class is asserted here.

2 Escape from a subspace

Lemma 2.1. *Let E be finite-dimensional over a field, $x \in E$, and $T_a \in \mathrm{End}(E)$. If an orbit vector $T_w x$ lies outside a subspace W, then one does so with $|w| \leq \dim W$.*

Proof. Let $L_j = \mathrm{span}\{T_w x : |w| < j\}$, so $L_0 = 0$ and $L_{j+1} = L_j + \sum_a T_a L_j$ for $j \geq 1$, with $L_1 = \mathrm{span}\{x\}$. If $L_j = L_{j+1}$, the span is invariant and contains the whole orbit (also for $j = 0$, when $x = 0$). Suppose every word of length at most $d = \dim W$ stays in W. Then

$L_{d+1} \subseteq W$. Because the orbit eventually escapes, none of the inclusions $L_j \subseteq L_{j+1}$, $0 \le j \le d$, can be an equality. Thus $\dim L_{d+1} \ge d+1$, a contradiction. $\square$

Write $N = n(n+1)/2$. The symmetric matrices $\mathrm{Sym}_n(\mathbb{R})$ have dimension N. On the lifted space $\mathbb{R} \oplus \mathrm{Sym}_n(\mathbb{R})$, let

$$\mathcal{T}_a(c, X) = (c, M_a X M_a^{\mathsf{T}}), \qquad \mathcal{T}_w(c, X) = (c, M_w X M_w^{\mathsf{T}}).$$

The constant coordinate makes affine matrix observations linear. Unlike the scalar observation in r2, an observation onto $\mathrm{Sym}_r(\mathbb{R})$ can have rank $r(r+1)/2$, leaving a smaller kernel from which to escape.

3 Finite compressed returns and rank loss

Lemma 3.1. *If $\mathcal{R} \subseteq M_r(\mathbb{R})$ is finite and multiplicatively closed, there is a positive-definite symmetric Q such that $KQK^{\mathsf{T}} = Q$ for every invertible $K \in \mathcal{R}$.*

Proof. If there are no invertible members, take $Q = I_r$. Otherwise the invertible members form a finite group G: equality between two positive powers yields identity and inverse within $\mathcal{R}$. Set $Q = \sum_{g \in G} g g^{\mathsf{T}}$. For nonzero v, every summand $v^{\mathsf{T}} g g^{\mathsf{T}} v = \|g^{\mathsf{T}} v\|^2$ is positive, so Q is positive definite. Left multiplication by a group member permutes the summands. Dimension zero is vacuous. $\square$

Fix a word h, put $H = M_h$, and let $r = \operatorname{rank} H$. A rank factorization gives $H = UV$, with U of size $n \times r$ and V of size $r \times n$. Define compressed returns

$$K_w = V M_w U, \qquad H M_w H = U K_w V.$$

They are finite and closed under multiplication, since $K_x K_y = K_{xhy}$. If $\operatorname{rank}(H M_w H) \ge r$, then K_w is invertible. This follows from $\operatorname{rank}(U K_w V) \le \operatorname{rank} K_w \le r$.

Lemma 3.2. *If some product has rank below $r = \operatorname{rank} M_h$, there is a word w such that*

$$|w| \le c_r := 1 + \frac{n(n+1)}{2} - \frac{r(r+1)}{2}, \qquad \operatorname{rank} M_{hwh} < r.$$

Proof. Here $r > 0$. If H^2 already has smaller rank, take $w = \epsilon$. Otherwise VU is invertible, so V has a right inverse $W = U(VU)^{-1}$. Choose Q as in Lemma 3.1 for the compressed returns, and use the seed $(1, UQU^{\mathsf{T}})$ in the lifted space. The linear map

$$F(c, X) = V X V^{\mathsf{T}} - cQ \colon \mathbb{R} \oplus \mathrm{Sym}_n(\mathbb{R}) \longrightarrow \mathrm{Sym}_r(\mathbb{R})$$

is onto: $F(0, WYW^{\mathsf{T}}) = Y$. Hence $\dim \ker F = c_r$. On the orbit of the seed,

$$F(\mathcal{T}_w(1, UQU^{\mathsf{T}})) = K_w Q K_w^{\mathsf{T}} - Q.$$

Let z have $\operatorname{rank} M_z < r$. Its defect cannot vanish: otherwise $r = \operatorname{rank} Q = \operatorname{rank}(K_z Q K_z^{\mathsf{T}}) \le \operatorname{rank} M_z < r$. Lemma 2.1 therefore gives a word w of length at most c_r with nonzero defect. An invertible K_w would have zero defect by construction. Thus K_w is singular and $\operatorname{rank} M_{hwh} < r$. $\square$

4 The exact descent budget

Start at the empty word of rank n and length zero. Replacing h by hwh costs $2|h| + |w|$. For $d \geq 0$, define

$$L_0 = 0, \qquad L_{d+1} = 2L_d + 1 + \frac{n(n+1)}{2} - \frac{(n-d)(n-d+1)}{2} \quad (0 \leq d < n).$$

These budgets are nondecreasing. Induction gives the additive identity

$$L_d + \frac{n(n+1)}{2} + (n-d) = n2^d + \frac{(n-d)(n-d+1)}{2} \quad (0 \leq d \leq n). \tag{2}$$

For completeness, replacing d by $d+1$, expanding L_{d+1}, and using $t(t+1)/2 = t(t-1)/2 + t$ with $t = n - d$ proves the induction step.

Proof of Theorem 1.1. At any rank $r > s$, apply Lemma 3.2. Maintain the invariant that the current word has length at most L_{n-r}. The lemma and recurrence bound the next length by L_{n-r+1}. If the new rank is $q < r - 1$, monotonicity allows the larger budget L_{n-q}; thus skipped ranks cause no problem. Rank strictly decreases until it equals s. Substitution of $d = n - s$ in (2) gives $L_{n-s} = B(n, s)$. The rational statement follows by embedding $\mathbb{Q}$ into $\mathbb{R}$: every word product is preserved, its rank is unchanged, and finiteness is preserved. The rank-zero case is mortality. Direct algebra gives (1). □

5 The sharp planar threshold

Theorem 5.1. *Every mortal finite monoid of real 2×2 matrices has a zero word of length at most four. The constant four is attained by an integer pair, hence is sharp over both $\mathbb{R}$ and $\mathbb{Q}$.*

Upper bound. Theorem 1.1 supplies a word of length at most $B(2, 0) = 5$. Suppose no zero word has length at most four, and write a five-letter zero product as $ABCDE = 0$. Every letter is nonzero. A nonzero singular planar matrix is rank one. If a middle letter, say $C = uv^{\mathsf{T}}$, is singular, the factorization $(ABu)(v^{\mathsf{T}}DE) = 0$ forces $ABC = 0$ or $CDE = 0$; similarly for B, D. This contradicts minimality, so B, C, D are invertible. Also E is singular, since otherwise $ABCD = 0$. Write $E = uv^{\mathsf{T}}$ with $u, v \neq 0$.

Average a positive quadratic form over the finite group of units of S, now in the vector-invariant convention $g^{\mathsf{T}}Qg = Q$. The four nonzero vectors

$$u, \quad Du, \quad CDu, \quad BCDu$$

lie on one positive ellipse. For any such unit g, gE is a nonzero rank-one element of a finite monoid, and $\operatorname{tr}(gE) = v^{\mathsf{T}}gu$ belongs to $\{0, 1, -1\}$. Indeed, for a rank-one matrix X, $X^k = (\operatorname{tr} X)^{k-1}X$ for $k \geq 1$; finite powers force its real trace to be zero or a real root of unity.

An ellipse intersected with the line $v^{\mathsf{T}}x = 0$ yields at most one projective direction. The intersections with $v^{\mathsf{T}}x = \pm 1$ yield at most two more: changing sign moves each point to the line $v^{\mathsf{T}}x = 1$, which meets the ellipse in at most two points. Thus two of the four vectors have the same direction. Delete the corresponding segment of the three-letter middle word. The remaining outside prefix still kills the proportional vector because $ABCDu = 0$. Multiplication by v^{T} produces a zero word of length at most four, a contradiction. □

Sharpness. Take

$$R = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Here $R^3 = I$, $P^2 = P$, $PRP = -P$, and $PR^2P = 0$. Every product lies in the finite set

$$\{I, R, R^2, 0\} \;\cup\; \{\pm R^i P R^j : 0 \le i, j < 3\}.$$

A zero word must contain at least two occurrences of P. Up to length three, adjacent occurrences collapse to P, while the only separated pair is $PRP = -P \neq 0$; products with at most one P are nonzero. Hence the threshold is exactly four. These matrices realize the classical Černý three-state example on its zero-sum plane, up to basis and generator convention [2]. □

The real plane action is irreducible: $R^2 + R + I = 0$, and $t^2 + t + 1$ has no real root, so R preserves no real line.

6 Invariant flags and sharp small factors

Put

$$f(d) = \begin{cases} 4, & d = 2, \\ B(d,0), & d \neq 2. \end{cases} \qquad \text{Thus } f(0) = 0,\ f(1) = 1.$$

Theorem 6.1. *Let a finite mortal monoid act on a finite-dimensional real vector space E. Suppose $0 = V_0 \subseteq V_1 \subseteq \cdots \subseteq V_k = E$ is an invariant flag, and put $d_i = \dim(V_i/V_{i-1})$. There is a zero word of length at most $\sum_{i=1}^{k} f(d_i)$. In particular, if there are k_1 factors of dimension one and k_2 factors of dimension two, the bound is $k_1 + 4k_2 \le 2\dim E$.*

Proof. The induced monoid on each quotient is finite and mortal. Apply Theorem 1.1 or Theorem 5.1 to obtain a word w_i, of length at most $f(d_i)$, mapping V_i into V_{i-1}. The concatenation $w_1 \cdots w_k$ sends E successively down the flag, since the rightmost factor acts first. Its length is at most the sum. Finally $\dim E = k_1 + 2k_2$. This is the invariant-subspace concatenation argument of Almeida–Steinberg [1, Lemma 2.4 and Proposition 2.5], applied to the present factor bounds. □

Proposition 6.2. *For every $k_1, k_2 \ge 0$, there is a finite integer block system in dimension $k_1 + 2k_2$ whose mortality threshold is exactly $k_1 + 4k_2$. Each block has dimension one or two, and each real block action is irreducible.*

Proof. Use k_1 one-dimensional blocks with a zero generator and k_2 copies of the pair R, P above. Each letter acts on only one designated block and as identity on all others. Projecting a word to the letters assigned to a block gives precisely that block's product. Therefore the word is zero exactly when all projections are zero; the total length is the sum of the projected lengths. The lower bounds one and four add, and concatenating the blockwise shortest words attains their sum. The monoid embeds in a finite product of the finite block monoids. One-dimensional actions are irreducible and the planar assertion follows from the preceding paragraph. When there are no blocks, the empty word realizes threshold zero. □

We state these results for an explicit invariant flag and explicit block systems. No general composition-series construction or Jordan–Hölder classification is needed for the claimed bounds.

7 A boundary for the finiteness hypothesis

Proposition 7.1. *For every $N \ge 0$, two real-linear operators on the plane generate a bounded mortal product monoid, each generator has a finite set of powers, and no word of length at most N is zero.*

Proof. Set $m = N + 1$, $\theta = \pi/(2m)$. Identify the real plane with $\mathbb{C}$. Let $R_\theta z = e^{i\theta} z$, and let $Pz = \operatorname{Re} z$. Both are contractions, so all products have operator norm at most one. Moreover $R_\theta^{4m} = I$, $P^2 = P$, and $PR_\theta^m P = 0$.

A word of length less than m, acting on 1 from right to left, always gives $qe^{ij\theta}$ with $q > 0$ and $0 \le j \le$ the number of letters so far. A rotation increments j; a projection replaces q by $q\cos(j\theta) > 0$ and resets j to zero. The angle remains below $\pi/2$. Such a word cannot be the zero operator, and every word of length at most N has length less than m. □

This proves failure of any dimension-only bound after replacing finite entire monoid by bounded monoid plus finite powers of the individual generators. It is not a claim that every element has finite powers. We use only the proved nonuniformity assertion; no exact threshold formula for this family is required.

8 Compressed witnesses

A straight-line program here is a finite directed acyclic list of gates: an empty-word gate, a letter gate, or a concatenation gate referring to two earlier gates. One gate is designated as the output. Shared references permit a word to have an exponentially long expansion.

Theorem 8.1. *Under Theorem 1.1's hypotheses, a word attaining minimum rank s, with expanded length at most $B(n, s)$, can be represented by a valid straight-line program with at most*

$$1 + n(2(1 + n(n + 1)/2) + 1) = n^3 + n^2 + 3n + 1$$

gates. This holds over both $\mathbb{R}$ and $\mathbb{Q}$. Evaluating its gates in a monoid gives the same product as evaluating its expanded word.

Proof. Start with one empty-word gate. At a rank-descent step retain the old output gate for h. For each letter of the short middle word w, add its input gate and a concatenation gate to extend the current prefix. One final gate concatenates the resulting hw with a reference to the *old* output h. This implements hwh using $2|w| + 1$ new gates, without duplicating the program for h. There are at most n strict rank decreases, and $|w| \le 1 + n(n + 1)/2$. The same descent proof preserves the expanded length $B(n, s)$. Gate evaluation correctness is induction on the acyclic list, using compatibility of word evaluation with concatenation. Scalar extension preserves ranks and leaves the word program unchanged. □

The formal representation lists later gates before their dependencies and uses validated relative references into the tail; this is the same acyclic construction in reverse storage order. The empty gate is counted. The theorem establishes a small witness, not a polynomial-time procedure to discover it. In particular, it does not bound the cost of finding the invariant forms, choosing middle words, or representing arbitrary real coefficients. Specialized zero–one monoids have stronger constructive results [3].

9 Formal verification and limits

The accompanying Lean 4.33.1 project [5], with pinned mathlib [9], uses the namespace `FiniteMonoidMortality`. Its single entry point is `Publication.lean`; `Audit.lean` checks endpoint statements and transitive axioms, and `Specification.lean` checks the advertised source statements after the documented renaming. The module guide and claim map identify the precise formal coverage.

The development formalizes the real and rational minimum-rank bounds, mortality specialization and comparison with r2; the planar upper bound and the explicit integer lower example; quotient actions and flag bounds; exact thresholds of the independent block construction and irreducibility of its planar component; the bounded plane counterexample; and valid compressed witness existence with evaluation correctness. The real block sharpness interpretation uses scalar extension of the explicitly formalized integer construction. The general representation-image interpretation in Section 1 and bibliographic priority comparisons are prose consequences or historical statements, not separate formal declarations.

The proof package is extracted at declaration level from shared research code. Its endpoint assumptions and definitions are checked against the pinned source, and the extracted project is rebuilt with an empty project build directory. The endpoint audits allow only Lean's standard propositional extensionality, classical choice and quotient soundness axioms; no project axiom or admitted proof supports these results. These checks concern the encoded propositions. They do not establish global optimality, historical novelty, or the deferred algorithmic claims.

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